

Probability & Statistics

Course Project Report

Bayesian View of E-Commerce Resellers

*A Probabilistic Framework for Evaluating Seller Trustworthiness
from Review Data*

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Abstract

This report addresses a common consumer decision problem in e-commerce: given three sellers offering the same product at the same price, with review counts of 10 (100% positive), 50 (96% positive), and 200 (93% positive) respectively, which seller should a rational buyer prefer? We formalise this as a Bayesian inference problem, modelling each seller's latent quality as a Beta-distributed random variable. We derive the closed-form expression for the probability that one Beta random variable dominates another and extend this to the three-way case. Our key finding is that the seller with 50 reviews (96% positive) has the highest probability of being the best choice (44.63%), narrowly ahead of the perfect-but-sparse 10-review seller (41.22%). We further demonstrate that expected-value rankings can be misleading when posterior variance is high, validating probability of dominance as the superior decision criterion.

Contents

1	Introduction and Motivation	2
2	The Statistical Model	2
2.1	Framing the Problem	2
2.2	The Likelihood: Binomial Distribution	3
3	The Beta Distribution	3
3.1	Deriving the Beta PDF from First Principles	3
3.2	Applying the Beta Distribution to the Seller Problem	4
4	Expectation and Variance of the Beta Distribution	4
4.1	Expected Value	4
4.2	Variance	5
5	Probability of Dominance Between Beta Random Variables	5
5.1	Pairwise Dominance: $P(X_1 > X_2)$	5
5.2	Three-Way Dominance	5
6	Results	6
6.1	Posterior Beta Distributions	6
6.2	Expected Values	6
6.3	Probability of Dominance (Main Result)	6
7	Why Expected Value Alone is Insufficient	7
8	R Implementation	8
8.1	Visualising the Three Posterior Beta Distributions	8
8.2	Computing Probability of Dominance (Three-Way)	8
8.3	Demonstrating the Variance Effect (Beta(5, 5) vs Beta(20, 6))	9
9	Further Applications of Comparing Beta Distributions	10
9.1	Machine Learning: Comparing Classifiers	11
9.2	Website Performance: A/B Testing Click-Through Rates	11
10	Conclusion	11
	Appendix: Summary of Key Formulae	12

1. Introduction and Motivation

Consider the following scenario: you are browsing for a product online and find three different sellers offering it at essentially the same price. Their review histories are:

Seller	Total Reviews	Positive	Rating
D1	10	10	100%
D2	50	48	96%
D3	200	186	93%

Which seller should you buy from?

Most people intuitively feel that more data gives more confidence in a rating, and that a 100% score from only 10 reviews is suspicious. Yet the naive approach of simply comparing percentages ($100\% > 96\% > 93\%$) would select D1. The question is: *what is the rational, mathematically principled way to reason about this trade-off between the confidence gained from more data versus a lower absolute percentage?*

This report answers that question using **Bayesian inference**. The key idea is to treat each seller's underlying quality—the long-run probability that a transaction goes well—as an unknown continuous random variable, and then use the observed review data to form a *posterior probability distribution* over that variable.

Topics Covered

- Probability Density Functions (PDFs)
- Binomial Distribution as the likelihood model
- Beta Distribution as the conjugate prior
- Bayesian Updating
- Expectation and Variance of the Beta distribution
- Probability of dominance between Beta random variables (pairwise and three-way)
- R implementation and visualisations
- Applications beyond e-commerce

2. The Statistical Model

2.1 Framing the Problem

Each seller–buyer interaction is modelled as a **Bernoulli trial**: the buyer either has a positive experience (success, with probability S) or a negative experience (failure, with probability $1 - S$). The seller's true quality $S \in [0, 1]$ is unknown to the buyer.

What we have: $P(\text{data} \mid S)$ — the probability of observing x successes in n trials given a known success rate S .

What we want: $P(S \mid \text{data})$ — the probability that the success rate is S , given the observed reviews.

This reversal is exactly what **Bayes' theorem** provides.

2.2 The Likelihood: Binomial Distribution

If the true success rate S were known, the probability of obtaining exactly x positive reviews out of n total reviews would follow the **Binomial distribution**:

$$P(X = x \mid S) = \binom{n}{x} S^x (1 - S)^{n-x} \quad (1)$$

where x is the number of successes and $n - x$ is the number of failures.

However, S is *not* known—it is the very quantity we wish to estimate. We need a probability distribution over the continuous variable S .

3. The Beta Distribution

3.1 Deriving the Beta PDF from First Principles

We seek a function $f(s)$ that describes our belief about S after observing a successes and b failures. Intuitively, this function should be proportional to $s^a(1 - s)^b$:

$$f(s) \propto s^a(1 - s)^b, \quad a, b \in \mathbb{Z}^+$$

Since S is a continuous random variable, $f(s)$ must be a proper probability density function and therefore must integrate to 1:

$$\int_0^1 f(s) ds = 1 \implies \int_0^1 c \cdot s^a(1 - s)^b ds = 1$$

This means $c = \frac{1}{\int_0^1 s^a(1 - s)^b ds}$.

The integral in the denominator is the **Beta function**:

$$B(a + 1, b + 1) = \frac{a! b!}{(a + b + 1)!} = \frac{1}{c}$$

Therefore the normalised density is:

$$f(s) = \frac{s^a(1 - s)^b}{B(a + 1, b + 1)}$$

Substituting $\alpha = a + 1$ and $\beta = b + 1$ (so that $\alpha - 1$ equals the number of successes and $\beta - 1$ equals the number of failures):

$$f(s; \alpha, \beta) = \frac{s^{\alpha-1}(1-s)^{\beta-1}}{B(\alpha, \beta)}, \quad s \in [0, 1], \quad \alpha, \beta > 0 \quad (2)$$

We write $S \sim \text{Beta}(\alpha, \beta)$.

The Beta distribution is the **conjugate prior** for the Binomial likelihood: starting with a $\text{Beta}(\alpha_0, \beta_0)$ prior and observing a successes and b failures yields a $\text{Beta}(\alpha_0 + a, \beta_0 + b)$ posterior. With a uniform (non-informative) prior $\text{Beta}(1, 1)$, the posterior is simply:

$$\text{Posterior} = \text{Beta}(a + 1, b + 1)$$

3.2 Applying the Beta Distribution to the Seller Problem

For each seller we compute the posterior parameters:

Seller	Total	Positive (a)	Negative (b)	$\alpha = a+1$	$\beta = b+1$	Posterior
D1	10	10	0	11	1	$\text{Beta}(11, 1)$
D2	50	48	2	49	3	$\text{Beta}(49, 3)$
D3	200	186	14	187	15	$\text{Beta}(187, 15)$

The corresponding posterior density functions are:

$$f(x_1) = \frac{x_1^{10}}{B(11, 1)}, \quad f(x_2) = \frac{x_2^{48}(1-x_2)^2}{B(49, 3)}, \quad f(x_3) = \frac{x_3^{186}(1-x_3)^{14}}{B(187, 15)}$$

4. Expectation and Variance of the Beta Distribution

4.1 Expected Value

$$E(X) = \int_0^1 x f(x) dx = \frac{1}{B(\alpha, \beta)} \int_0^1 x^\alpha (1-x)^{\beta-1} dx = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)}$$

Using the identity $B(\alpha+1, \beta) = \frac{\alpha! (\beta-1)!}{(\alpha+\beta)!}$:

$$E(X) = \frac{\alpha}{\alpha + \beta} \quad (3)$$

Applying to the three sellers:

$$E(X_1) = \frac{11}{12} \approx 0.9167, \quad E(X_2) = \frac{49}{52} \approx 0.9423, \quad E(X_3) = \frac{187}{202} \approx 0.9257$$

By expected value alone: $E(X_2) > E(X_3) > E(X_1)$, suggesting D2 is the best choice.

4.2 Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{B(\alpha + 2, \beta)}{B(\alpha, \beta)} - \left(\frac{\alpha}{\alpha + \beta} \right)^2$$

Simplifying:

$$\text{Var}(X) = \frac{\alpha(\alpha + 1)}{(\alpha + \beta + 1)(\alpha + \beta)} - \frac{\alpha^2}{(\alpha + \beta)^2}$$

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2} \quad (4)$$

As the total number of reviews grows, $\alpha + \beta$ increases and the variance decreases, meaning the posterior distribution tightens around the true quality.

5. Probability of Dominance Between Beta Random Variables

5.1 Pairwise Dominance: $P(X_1 > X_2)$

Let $X_1 \sim \text{Beta}(\alpha_1, \beta_1)$ and $X_2 \sim \text{Beta}(\alpha_2, \beta_2)$ be independent. We wish to compute:

$$P(X_1 > X_2) = \int_0^1 \int_0^{x_1} \frac{x_1^{\alpha_1-1}(1-x_1)^{\beta_1-1}}{B(\alpha_1, \beta_1)} \cdot \frac{x_2^{\alpha_2-1}(1-x_2)^{\beta_2-1}}{B(\alpha_2, \beta_2)} dx_2 dx_1$$

Evaluating the inner integral and applying integration by parts via the **regularised incomplete Beta function** $I_{x_1}(\alpha_2, \beta_2) = \int_0^{x_1} \frac{x_2^{\alpha_2-1}(1-x_2)^{\beta_2-1}}{B(\alpha_2, \beta_2)} dx_2$:

$$I_{x_1}(\alpha_2, \beta_2) = 1 - (1-x_1)^{\beta_2} - \sum_{j=1}^{\alpha_2-1} \frac{x_1^j (1-x_1)^{\beta_2}}{(\alpha_2-j) B(\alpha_2-j, \beta_2)}$$

After repeated application of integration by parts and simplification:

$$P(X_1 > X_2) = 1 - \sum_{i=0}^{\alpha_2-1} \frac{B(\alpha_1+i, \beta_1+\beta_2)}{(\beta_2+i) B(1+i, \beta_2) B(\alpha_1, \beta_1)} \quad (5)$$

5.2 Three-Way Dominance

To determine which seller is best, we compute the probability that seller k beats *both* other sellers simultaneously. For all three combinations:

$$P(X_3 > \max\{X_1, X_2\}) = 1 - P(X_1 > X_3) - P(X_2 > X_3) + T_3$$

$$P(X_2 > \max\{X_1, X_3\}) = 1 - P(X_1 > X_2) - P(X_3 > X_2) + T_2$$

$$P(X_1 > \max\{X_2, X_3\}) = 1 - P(X_1 > X_2) - P(X_2 > X_3) + T_1$$

where the correction terms (derived from the triple overlap) are:

$$T_3 = \sum_{i=0}^{\alpha_1-1} \sum_{j=0}^{\alpha_2-1} \frac{B(\alpha_3 + i + j, \beta_3 + \beta_2 + \beta_1)}{(\beta_1 + i)(\beta_2 + j) B(1 + i, \beta_1) B(1 + j, \beta_2) B(\alpha_3, \beta_3)} \quad (6)$$

$$T_2 = \sum_{i=0}^{\alpha_1-1} \sum_{j=0}^{\alpha_3-1} \frac{B(\alpha_2 + i + j, \beta_2 + \beta_3 + \beta_1)}{(\beta_1 + i)(\beta_3 + j) B(1 + i, \beta_1) B(1 + j, \beta_3) B(\alpha_2, \beta_2)} \quad (7)$$

$$T_1 = \sum_{i=0}^{\alpha_2-1} \sum_{j=0}^{\alpha_3-1} \frac{B(\alpha_1 + i + j, \beta_1 + \beta_3 + \beta_2)}{(\beta_2 + i)(\beta_3 + j) B(1 + i, \beta_2) B(1 + j, \beta_3) B(\alpha_1, \beta_1)} \quad (8)$$

6. Results

6.1 Posterior Beta Distributions

Figure 1 shows the three posterior distributions. Note how Beta(11, 1) (D1) is extremely wide—reflecting high uncertainty from only 10 reviews—while Beta(187, 15) (D3) is sharply concentrated near 0.925, reflecting high confidence from 200 reviews.

Beta Distributions: Beta(11, 1) *vs.* Beta(49, 3) *vs.* Beta(187, 15)

The three curves represent the posterior distribution over the seller's true quality S . A wider curve means more uncertainty; a narrower, taller curve means the data strongly constrains the estimate. Sellers with more reviews produce tighter posteriors even when their percentage rating is slightly lower.

Figure 1: Posterior Beta distributions for the three sellers. Blue: D1 Beta(11, 1); Red: D2 Beta(49, 3); Violet: D3 Beta(187, 15).

6.2 Expected Values

Seller	Posterior	$E[X] = \alpha/(\alpha + \beta)$	Value	Rank
D1	Beta(11, 1)	11/12	0.9167	3rd
D2	Beta(49, 3)	49/52	0.9423	1st
D3	Beta(187, 15)	187/202	0.9257	2nd

6.3 Probability of Dominance (Main Result)

Applying equations (5)–(8) with $(\alpha_1, \beta_1) = (11, 1)$, $(\alpha_2, \beta_2) = (49, 3)$, $(\alpha_3, \beta_3) = (187, 15)$:

Seller	$P(\text{seller} = \text{best})$	Rank
D1 — Beta(11, 1)	0.4122	2nd
D2 — Beta(49, 3)	0.4463	1st
D3 — Beta(187, 15)	0.1415	3rd
<i>Sum</i>	1.0000	

Key insight: The seller with 50 reviews and 96% positive rating (D2) has the highest probability of being the best option: **44.63%**. Counterintuitively, the 100%-rated seller with only 10 reviews (D1) has a substantial 41.22% probability of being the best—nearly matching D2—because its wide posterior preserves significant probability mass at very high quality values. D3, despite having the most data, is ranked last because its posterior mean (0.9257) is the lowest of the three, and its narrow distribution does not compensate for this.

Recommendation: Purchase from **Seller D2**.

7. Why Expected Value Alone is Insufficient

The expected value rankings ($E(X_2) > E(X_3) > E(X_1)$) agree with the dominance rankings in *this particular example*. However, comparing means can be misleading when posterior variances differ substantially. Consider two hypothetical distributions:

$$X_1 \sim \text{Beta}(5, 5), \quad X_2 \sim \text{Beta}(20, 6)$$

Expected values:

$$E(X_1) = \frac{5}{10} = 0.500, \quad E(X_2) = \frac{20}{26} = 0.769$$

By this metric, X_2 is clearly better.

Variances:

$$\begin{aligned} \text{Var}(X_1) &= \frac{5 \times 5}{11 \times 100} = \frac{25}{1100} \approx 0.0227 \\ \text{Var}(X_2) &= \frac{20 \times 6}{27 \times 676} = \frac{120}{18252} \approx 0.00657 \end{aligned}$$

Note that $\text{Var}(X_1) > \text{Var}(X_2)$: the Beta(5, 5) distribution is much more spread out, with a non-trivial probability mass at values close to 1.

Probability of dominance: Using equation (5):

$$P(X_1 > X_2) = 0.9402, \quad P(X_2 > X_1) = 0.0598$$

Despite X_2 having a far higher expected value (0.769 vs. 0.500), the wide-variance distribution $X_1 \sim \text{Beta}(5, 5)$ has a **94.02%** probability of producing a higher value on any single draw.

This demonstrates that in decisions involving a single draw from each distribution (as in choosing one seller), comparing expected values is insufficient. **Probability of dominance is the correct criterion.**

8. R Implementation

8.1 Visualising the Three Posterior Beta Distributions

Listing 1: Plotting posterior Beta distributions for three sellers

```

1 x <- seq(0, 1, length = 10000)
2
3 y1 <- dbeta(x, shape1 = 11, shape2 = 1)
4 y2 <- dbeta(x, shape1 = 49, shape2 = 3)
5 y3 <- dbeta(x, shape1 = 187, shape2 = 15)
6
7 plot(x, y1, type = "l", col = "blue", lwd = 2,
8      ylim = c(0, max(y1, y2, y3)),
9      main = "Beta Distributions",
10     xlab = "x", ylab = "Density")
11 lines(x, y2, col = "red", lwd = 2)
12 lines(x, y3, col = "violet", lwd = 2)
13
14 legend("topleft",
15       legend = c("Beta(11,1)", "Beta(49,3)", "Beta(187,15)"),
16       col = c("blue", "red", "violet"), lwd = 2)

```

8.2 Computing Probability of Dominance (Three-Way)

Listing 2: Three-way dominance probability using closed-form summation. Requires the VGAM package for `lbeta()`.

```

1 library(VGAM)
2
3 # P(X beats both Y and Z) using the closed-form triple sum
4 P_X_beats_Y_and_Z <- function(aX, bX, aY, bY, aZ, bZ) {
5   total <- 0.0
6   for (i in 0:(aY - 1)) {
7     for (j in 0:(aZ - 1)) {
8       total <- total + exp(
9         lbeta(aX + i + j, bX + bY + bZ)
10        - log(bY + i)
11        - log(bZ + j)
12        - lbeta(1 + i, bY)
13        - lbeta(1 + j, bZ)
14        - lbeta(aX, bX)
15      )
16   }

```

```

17 }
18 return(1 - P_X_beats_Y(aX, bX, aY, bY)
19         - P_X_beats_Y(aX, bX, aZ, bZ) + total)
20 }
21
22 # P(X1 > X2) pairwise
23 P_X_beats_Y <- function(aX, bX, aY, bY) {
24   total <- 0.0
25   for (i in 0:(aY - 1)) {
26     total <- total + exp(
27       lbeta(aX + i, bX + bY)
28       - log(bY + i)
29       - lbeta(1 + i, bY)
30       - lbeta(aX, bX)
31     )
32   }
33   return(total)
34 }
35
36 run_probabilities <- function() {
37   a1 <- 11; b1 <- 1
38   a2 <- 49; b2 <- 3
39   a3 <- 187; b3 <- 15
40
41   P_d3_beats_d1_and_d2 <- P_X_beats_Y_and_Z(a3, b3, a1, b1, a2,
42     b2)
43   P_d2_beats_d1_and_d3 <- P_X_beats_Y_and_Z(a2, b2, a1, b1, a3,
44     b3)
45   P_d1_beats_d2_and_d3 <- P_X_beats_Y_and_Z(a1, b1, a2, b2, a3,
46     b3)
47
48   cat(sprintf("P(D3 beats D1 and D2): %.4f\n", P_d3_beats_d1_
49     and_d2))
50   cat(sprintf("P(D2 beats D1 and D3): %.4f\n", P_d2_beats_d1_
51     and_d3))
52   cat(sprintf("P(D1 beats D2 and D3): %.4f\n", P_d1_beats_d2_
53     and_d3))
54 }
55
56 run_probabilities()

```

Output:

Probability that D3 beats both D1 and D2: 0.1415
 Probability that D2 beats both D1 and D3: 0.4463
 Probability that D1 beats both D2 and D3: 0.4122

8.3 Demonstrating the Variance Effect (Beta(5,5) vs Beta(20,6))

Listing 3: Pairwise dominance: high-variance vs high-mean distribution

```

1 x <- seq(0, 1, length = 100)

```

```

2 y1      <- dbeta(x, shape1 = 5,  shape2 = 5)
3 y2      <- dbeta(x, shape1 = 20, shape2 = 6)
4 mean1   <- 5 / (5 + 5)
5 mean2   <- 20 / (20 + 6)
6
7 plot(x, y1, type = "l", col = "blue", lwd = 2,
8      ylim = c(0, max(y1, y2)),
9      main = "Beta Distributions with Expectation Values",
10     xlab = "x", ylab = "Density")
11 lines(x, y2, col = "red", lwd = 2)
12 abline(v = mean1, col = "blue", lty = 2)
13 abline(v = mean2, col = "red", lty = 2)
14 text(mean1, 2, labels = paste0("E(X) = ", round(mean1, 3)), pos
15      = 3, col = "blue")
16 text(mean2, 2, labels = paste0("E(X) = ", round(mean2, 3)), pos
17      = 3, col = "red")
18 legend("topleft", legend = c("Beta(5,5)", "Beta(20,6)"),
19      col = c("blue", "red"), lwd = 2)
20
21 # Pairwise dominance
22 P_X_beats_Y <- function(aX, bX, aY, bY) {
23   total <- 0.0
24   for (i in 0:(aY - 1)) {
25     total <- total + exp(lbeta(aX + i, bX + bY)
26                        - log(bY + i)
27                        - lbeta(1 + i, bY)
28                        - lbeta(aX, bX))
29   }
30   return(total)
31 }
32
33 P_d1_beats_d2 <- P_X_beats_Y(5, 5, 20, 6)
34 P_d2_beats_d1 <- P_X_beats_Y(20, 6, 5, 5)
35 cat(sprintf("P(Beta(5,5) beats Beta(20,6)): %.4f\n", P_d1_beats
36 _d2))
37 cat(sprintf("P(Beta(20,6) beats Beta(5,5)): %.4f\n", P_d2_beats
38 _d1))

```

Output:

Probability that d1 beats d2: 0.9402

Probability that d2 beats d1: 0.0598

9. Further Applications of Comparing Beta Distributions

The Beta–Binomial Bayesian framework developed here extends naturally to any domain where binary outcomes must be compared across groups with unequal sample sizes.

9.1 Machine Learning: Comparing Classifiers

In Bayesian machine learning, Beta distributions serve as conjugate priors for Binomial likelihood when estimating the probability of a binary event. Suppose two classifiers are evaluated:

- **Classifier A:** correctly classifies 85 out of 100 examples, modelled as $\text{Beta}(86, 16)$.
- **Classifier B:** correctly classifies 30 out of 50 examples, modelled as $\text{Beta}(31, 21)$.

By comparing Beta distributions rather than raw accuracy (85% vs. 60%), one can compute the probability that Classifier A will outperform Classifier B on future classifications, properly accounting for the uncertainty introduced by the smaller evaluation set of Classifier B.

9.2 Website Performance: A/B Testing Click-Through Rates

When comparing two website designs for user engagement:

- **Version A:** 1000 views, 100 clicks, modelled as $\text{Beta}(101, 901)$.
- **Version B:** 200 views, 30 clicks, modelled as $\text{Beta}(31, 171)$.

Even though Version B has a higher raw CTR (15% vs. 10%), computing the probability of dominance between the two Beta distributions gives a principled estimate of the likelihood that Version B truly outperforms Version A, taking into account the much larger sample behind Version A's estimate.

10. Conclusion

This project has demonstrated a complete Bayesian pipeline for a common e-commerce decision problem. Starting from the observation that review percentages alone are insufficient for rational decision-making, we:

1. Formalised the seller quality as a Beta-distributed posterior, derived as the conjugate prior for Binomial review data.
2. Derived closed-form expressions for the expectation ($\alpha/(\alpha + \beta)$) and variance ($\alpha\beta/[(\alpha + \beta + 1)(\alpha + \beta)^2]$) of the Beta distribution.
3. Derived a closed-form summation formula for $P(X_1 > X_2)$ using the regularised incomplete Beta function, and extended it to the three-way dominance case.
4. Showed that the 50-review seller (D2, 96% positive) has the highest probability of being the best choice (44.63%), while the 200-review seller (D3, 93% positive) is ranked last despite having the most data.
5. Demonstrated that expected-value comparisons can be misleading when posterior variances differ, and that probability of dominance is the correct decision criterion for single-draw problems.

The framework generalises to any domain involving binary outcomes with unequal sample sizes, including A/B testing, classifier comparison, and clinical trial analysis.

Appendix: Summary of Key Formulae

Quantity	Formula
Binomial likelihood	$P(X = x S) = \binom{n}{x} S^x (1 - S)^{n-x}$
Beta PDF	$f(s; \alpha, \beta) = \frac{s^{\alpha-1} (1-s)^{\beta-1}}{B(\alpha, \beta)}$
Posterior update	Prior Beta(1, 1) + data (a, b) $\Rightarrow$ Posterior Beta(a+1, b+1)
Expectation	$E(X) = \frac{\alpha}{\alpha + \beta}$
Variance	$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}$
Pairwise dominance	$P(X_1 > X_2) = 1 - \sum_{i=0}^{\alpha_2-1} \frac{B(\alpha_1 + i, \beta_1 + \beta_2)}{(\beta_2 + i) B(1 + i, \beta_2) B(\alpha_1, \beta_1)}$